

Summary Data on Acid Cropland and Lime Requirements in Tanzania

GAIA

www.acidsoils.africa

Last Updated: November 4, 2024

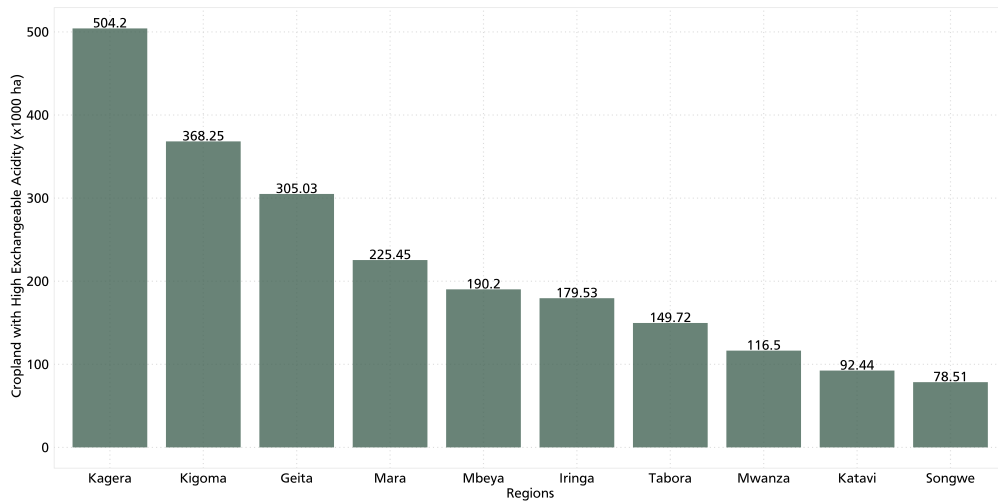
Outlook

- 1 Introduction
- 2 Acidic Cropland
- 3 Yield Loss
- 4 Lime Requirement
- 5 Yield Response
- 6 Additional Production
- 7 ROI for Lime Application
- 8 Profitability of Lime Application

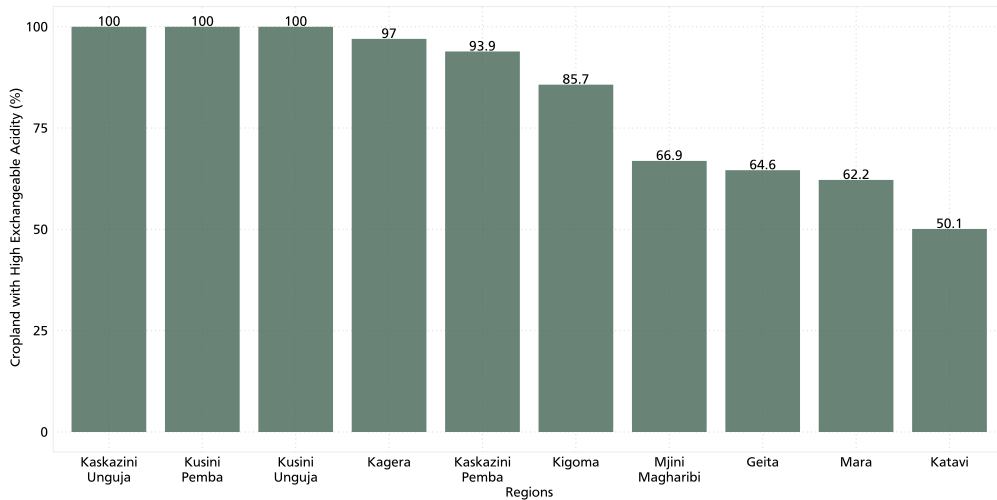
Introduction

- The slide deck provides a **data-driven, ex-ante analysis** of soil acidity across **Tanzania**, evaluating the scale of affected areas, potential crop yield impacts, and the expected benefits of lime application for proactive soil remediation.
- **Highlights:**
 - **Acidic Cropland:** Identifies regions with high acidity levels impacting crop yields.
 - **Yield Loss:** Quantifies productivity losses in key crops due to acidic soils.
 - **Lime Requirements:** Estimates the lime amounts needed to improve soil health in these areas.
 - **Economic Potential:** Highlights estimated increases in production, ROI and profitability from lime application.
- Provide policymakers, farmers, and stakeholders with actionable insights to prioritize areas with high potential for productivity gains and economic return from soil remediation.

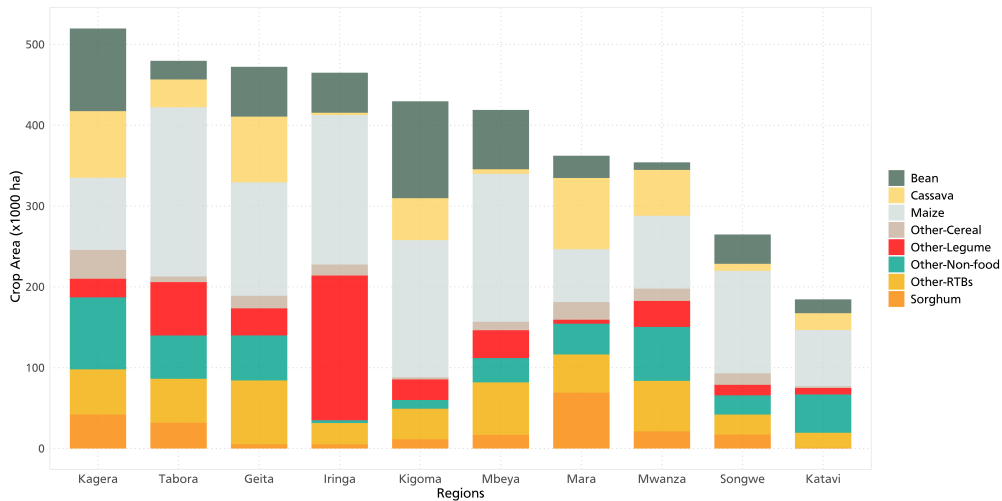
Top 10 Regions in Tanzania with the highest acidic cropland



Top 10 Regions in Tanzania with the highest acidic cropland (Percentage of total cropland)



Total crop area in the top-10 most acidic Regions, not segregated by acidic cropland



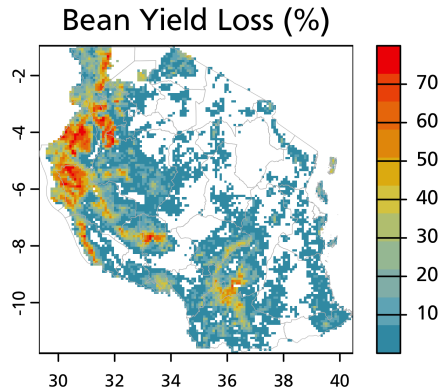
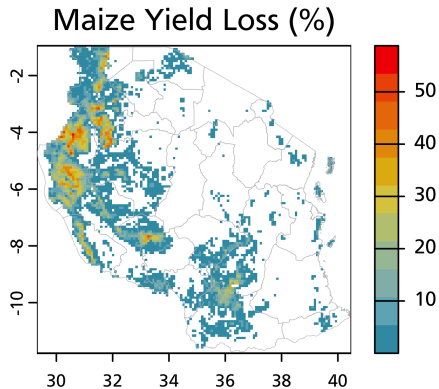
└ Total crop area distribution (ha)

└ Total crop area in the top-10 most acidic Regions, not segregated by acidic cropland

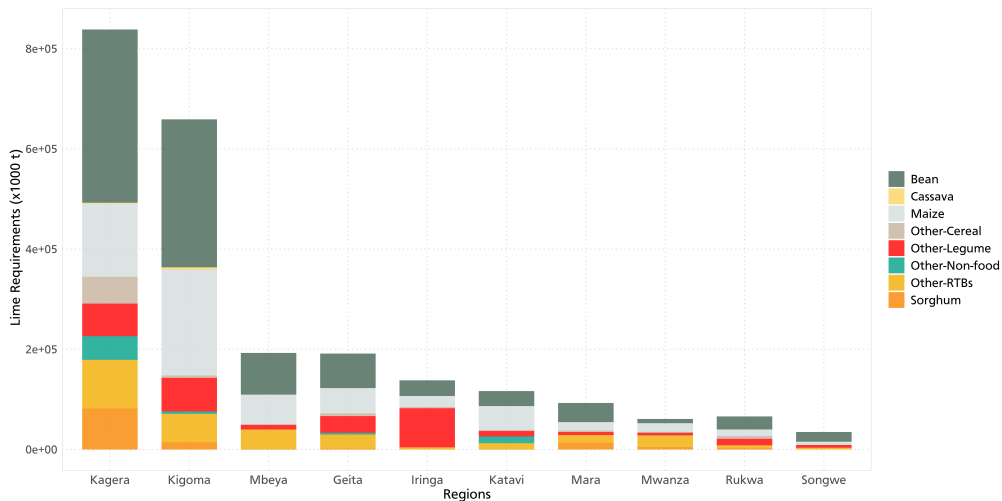
Total crop area in the top-10 most acidic Regions, not segregated by acidic cropland (table)

Regions	Bean	Cassava	Maize	Other-Cereal	Other-Legume	Other-Non-food	Other-RTBs	Sorghum
Geita	61.62	81.38	140.31	15.57	33.46	55.71	79.03	5.21
Iringa	49.55	2.95	184.81	13.70	179.01	3.50	26.34	5.18
Kagera	102.26	82.45	89.30	35.70	22.96	89.08	55.95	42.02
Katavi	17.08	20.77	69.70	2.14	8.02	47.37	18.60	0.89
Kigoma	119.90	51.83	169.76	2.62	25.52	10.77	37.86	11.38
Mara	27.52	88.24	65.43	21.82	4.85	38.03	47.33	69.10
Mbeya	73.37	5.62	183.11	10.51	34.35	30.19	65.01	16.78
Mwanza	9.36	56.74	90.11	15.23	32.22	66.68	62.60	21.13
Songwe	36.22	8.65	126.95	14.15	13.00	23.79	24.90	17.15
Tabora	23.04	34.40	209.44	6.93	66.03	53.56	54.51	31.78

Yield loss due to soil acidity for Maize and Bean in Tanzania



Lime requirements (t) for acid soil remediation in Tanzania

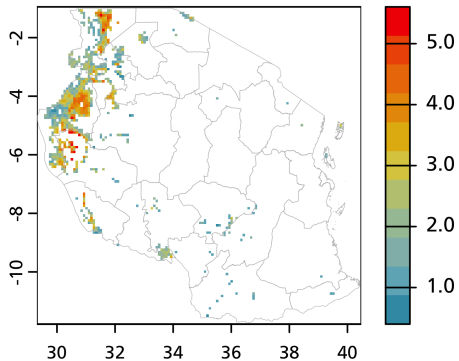


Lime requirements (1000 t) for acid soil remediation in Tanzania (table)

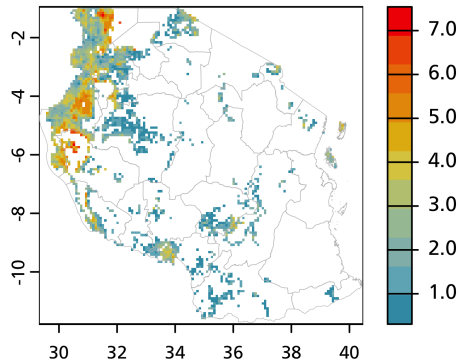
Regions	Bean	Cassava	Maize	Other-Cereal	Other-Legume	Other-Non-food	Other-RTBs	Sorghum
Geita	68.90	0.25	50.02	5.59	33.40	3.53	28.10	1.78
Iringa	31.07	-	22.10	2.44	77.93	-	4.20	0.27
Kagera	345.89	1.77	145.85	53.69	64.93	47.33	97.37	81.60
Katavi	29.91	0.14	47.41	2.38	10.74	13.50	12.00	0.60
Kigoma	295.41	4.52	211.39	4.73	67.14	4.67	56.71	14.54
Mara	38.12	-	16.61	3.23	6.10	-	15.38	13.42
Mbeya	83.24	-	59.17	1.32	9.13	0.02	38.97	0.96
Mwanza	8.43	-	17.33	1.83	5.06	0.23	23.33	4.79
Rukwa	26.08	-	12.82	5.88	12.86	-	6.08	2.35
Songwe	19.37	-	6.32	0.25	5.53	-	3.42	0.08

Lime rate for Maize and Bean in Tanzania

Maize Lime Rate (t/ha)

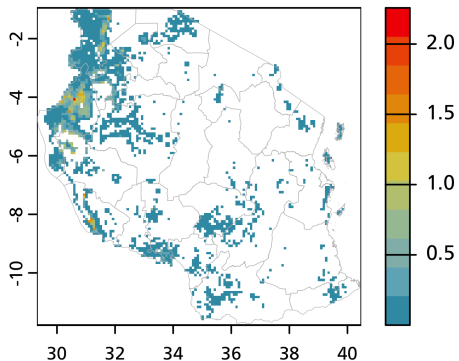


Bean Lime Rate (t/ha)

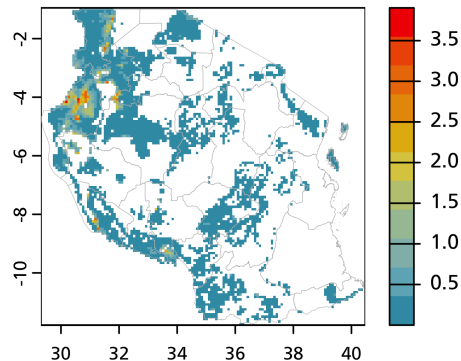


Yield response to lime application in Tanzania

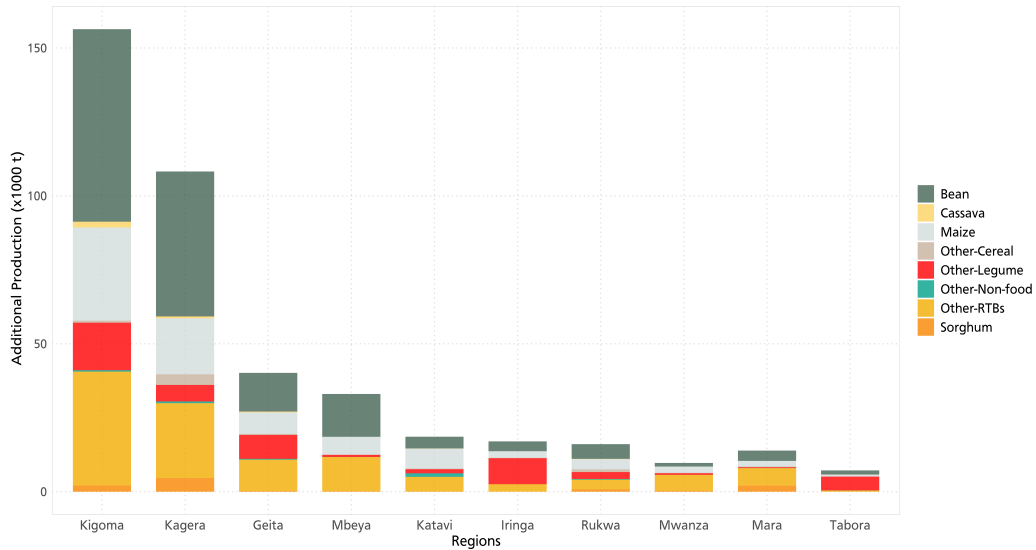
Maize Yield Response (t/ha)



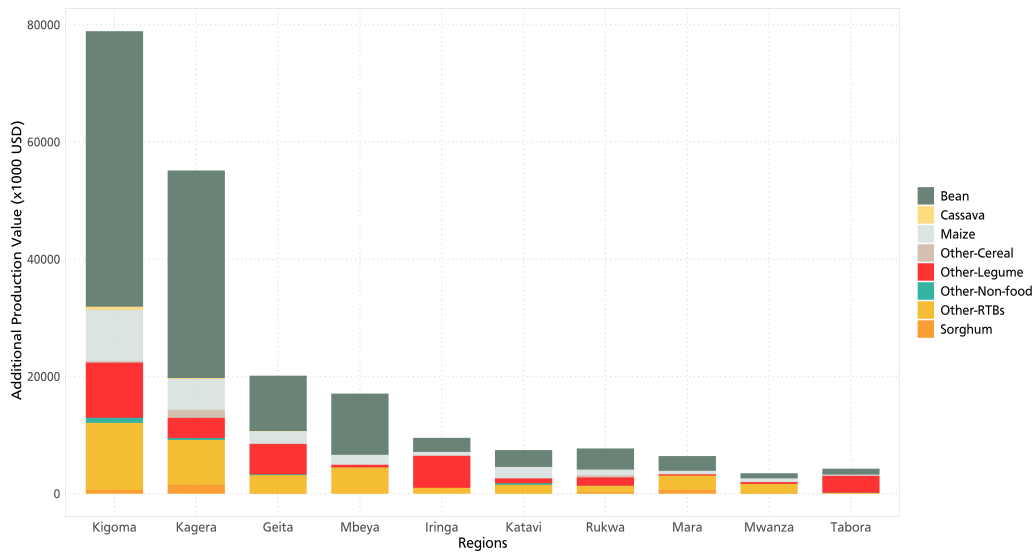
Bean Yield Response (t/ha)



Potential for additional production (tone) in top 10 acidic Regions in Tanzania

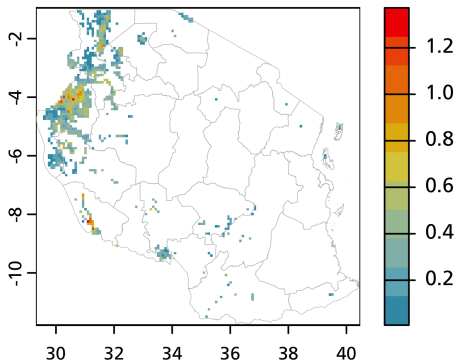


Potential for additional production value (US\$) in top 10 acidic Regions in Tanzania

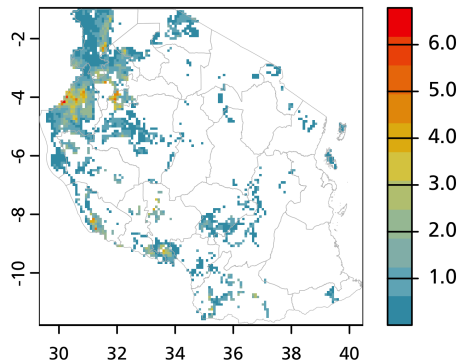


Return on investment for lime application in Tanzania

Maize ROI (USD/USD)



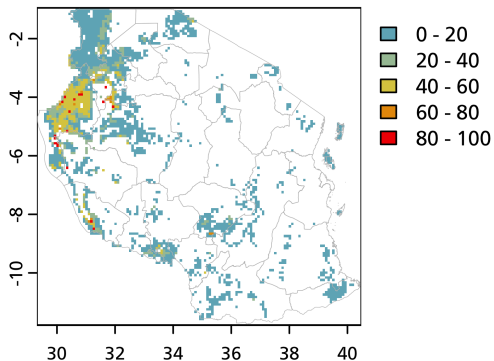
Bean ROI (USD/USD)



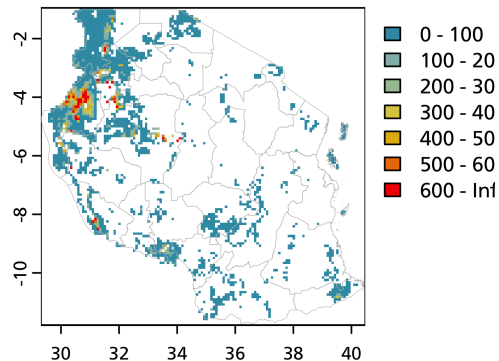
- └ Profitability of Lime Application
- └ Profitability of lime application in Tanzania

Profitability of lime application in Tanzania

Profitable area (%)



Weighted profit (USD/ha)



Main assumptions and parameters

- Acidic cropland is defined as having acidity saturation levels above 10% of the Effective Cation Exchange Capacity (ECEC).
- Crop yields are unaffected by soil acidity until a specific acidity saturation threshold Target Acidity Saturation(TAS) is reached; beyond this threshold, yields decrease linearly, eventually reaching zero at high acidity levels.
- The TAS that separates maximum yield from yield decline is used as a critical value for making liming recommendations.
- Actual yields in acid tropical soils are a fraction of the potential yields in non-acidic soils, determined by the estimated yield loss due to acidity.
- Economic benefits from liming are calculated by multiplying the additional yield after soil remediation by median crop prices from sub-Saharan Africa (2016-2020 FAOSTAT data). The cost of liming is calculated using estimated lime application rates and a fixed lime price of 100 US\$/ton.
- The benefits of liming are assessed only for the year of application due to a lack of robust, large-scale data on long-term effects in sub-Saharan Africa.

For details, visit www.acidsoils.africa.